Title

How can I create directories on the web server and correctly assign owners and permissions in Linux?

Service Request Summary

The client contacted IT support because they needed to create new directories on the web server under /var/www/website to add content to the site. When attempting to do so via the command line, they were unsure which commands to use to create the directories and how to correctly assign the required ownership and permissions for the server to function properly.

Troubleshooting Procedure

-The client may not be familiar with Linux commands for managing files and directories.
-The client might not know the correct syntax to change ownership and permissions.
-Administrative privileges (sudo) may also be required to apply the changes.

Identified Cause

It was determined that the client did not have prior experience using the necessary Linux commands to create directories and properly configure the ownership and permissions required by the web server.

Action Plan

- **Connect to the server via SSH using:**

```
ssh centos@localhost
```

```
PS C:\Users\AndreaLlorentePortil> ssh centos@localhost
```

- **Create the required directories on the web server:**

```
sudo mkdir -p /var/www/website/terms-and-conditions
sudo mkdir -p /var/www/website/about/our-ceo
```

** You can verify if the command executed successfully with::* **ls -l /var/www/website/**

```
[centos@localhost website]$ sudo mkdir -p /var/www/website/terms-and-con
ditions
[centos@localhost website]$ sudo mkdir -p /var/www/website/about/our-ceo
[centos@localhost website]$ ls -l /var/www/website/
```

- **Assign the correct owners and groups:**

```
sudo chown -R apache:apache /var/www/website
sudo chown ceo:apache /var/www/website/about/our-ceo
```

```
[centos@localhost website]$ sudo chown -R apache:apache /var/www/website
[centos@localhost website]$ sudo chown ceo:apache /var/www/website/about
/our-ceo
```

- **Set the required permissions:**

```
sudo chmod 755 /var/www/website/terms-and-conditions
sudo chmod 755 /var/www/website/about
sudo chmod 755 /var/www/website/about/our-ceo
```

```
[centos@localhost website]$ sudo chmod 755 /var/www/website/terms-and-co
nditions
[centos@localhost website]$ sudo chmod 755 /var/www/website/about
[centos@localhost website]$ sudo chmod 755 /var/www/website/about/our-ce
o
```

- **Verify that the changes were applied correctly with:**

```
ls -lR /var/www/website
```

```
[centos@localhost website]$ ls -lR /var/www/website
```

- **Confirm that directories, ownership, and permissions match the requirements.**

Client Response

The client confirmed that after following the instructions, they were able to create the directories and correctly apply ownership and permissions. They indicated that the step-by-step explanation helped them better understand how to use Linux commands.

Resources Used

- Linux command documentation: mkdir, chown, chmod, ls
- Internal web server configuration guide
- **Notes from the Linux system administration course lab**